

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

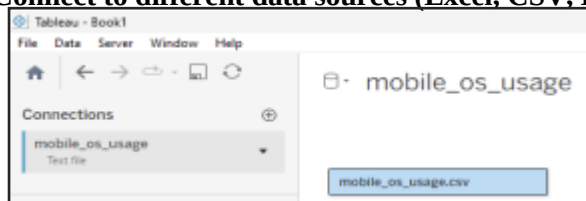
Practical – II

Roll No.: T032	Name: KHUSHI
Class: TYIT	Batch: 2
Date of Assignment: 31-07-26	Date/Time of Submission: 06-08-26

Part A – Program:

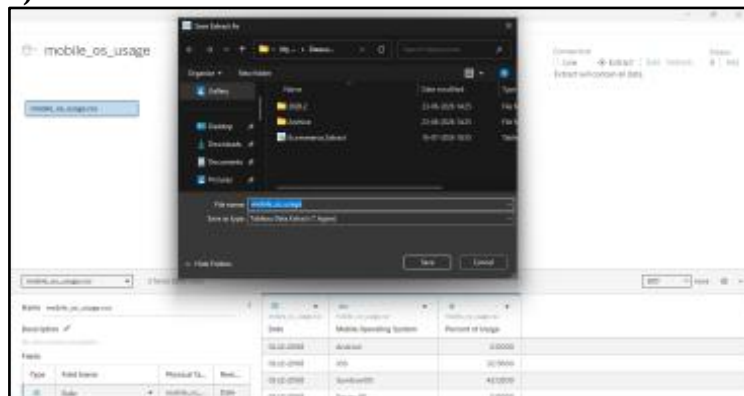
Working with Data Sources and Data Preparation in Tableau:

a) Connect to different data sources (Excel, CSV, Database)



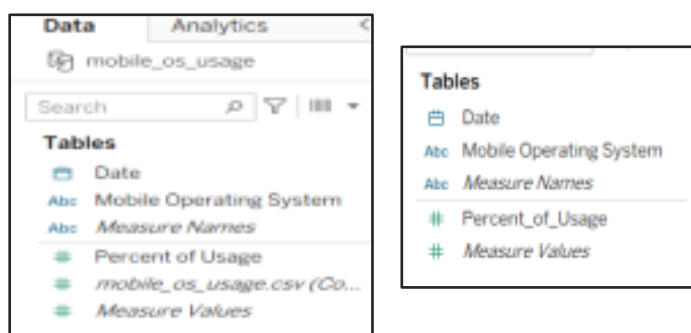
- Open Tableau.
- Select the required data source (Excel/CSV/Database).
- Browse and connect the file or database.
- Load the data into Tableau.

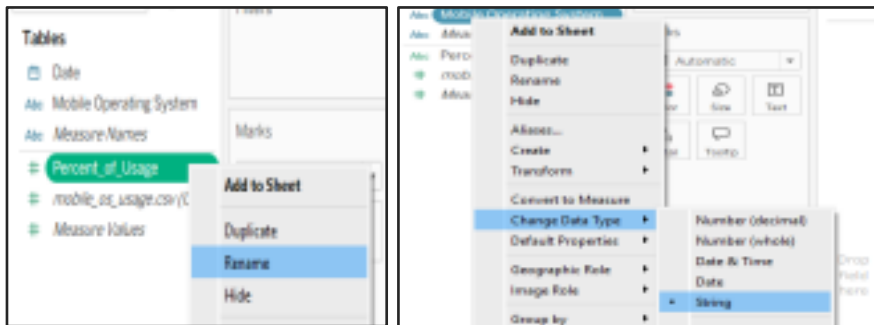
b) Create and use data extracts vs live connections:



- Connect to a data source.
- Choose **Extract** or **Live** connection.
- Save the extract if selected.
- Open a worksheet and compare performance.

c) Manage metadata (rename, hide, change data types):





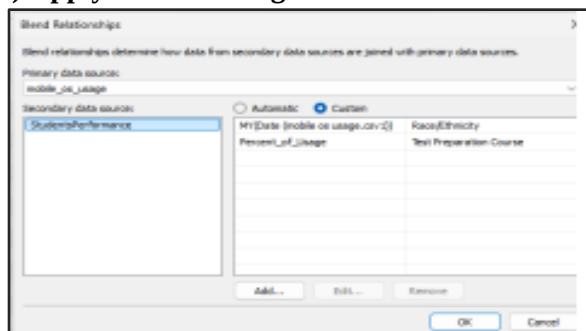
- Open the Data pane.
- Rename fields.
- Hide unnecessary columns.
- Change data types where required.

d) Perform joins (inner, left, right) on multiple tables:



- Add multiple tables.
- Drag one table onto another.
- Select the join type.
- Define the matching key.

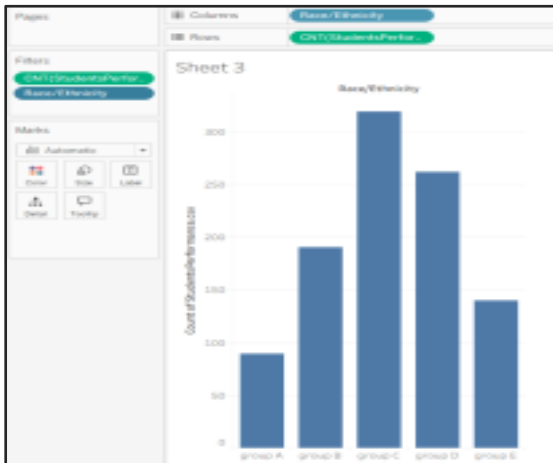
e) Apply data blending between two sources



- Connect to two data sources.
- Set one as Primary and the other as Secondary.

- Link common fields.
- Use fields from both sources in the visualization.

f) Implement different types of filters (dimension, measure, date)



- Drag a field to the Filters shelf.
- Select the filter type.
- Apply the required filter values.
- Display the filter if needed.

g) optimize performance using extracts and filtering.

Model	Type	Platform	Release Date	Status	Usage
Galaxy S21	Smartphone	Android	2021-01-01	Active	1000000000
Galaxy S20	Smartphone	Android	2020-01-01	Active	800000000
Galaxy S19	Smartphone	Android	2019-01-01	Active	600000000
Galaxy S18	Smartphone	Android	2018-01-01	Active	400000000
Galaxy S17	Smartphone	Android	2017-01-01	Active	200000000
Galaxy S16	Smartphone	Android	2016-01-01	Active	100000000
Galaxy S15	Smartphone	Android	2015-01-01	Active	50000000
Galaxy S14	Smartphone	Android	2014-01-01	Active	20000000
Galaxy S13	Smartphone	Android	2013-01-01	Active	10000000
Galaxy S12	Smartphone	Android	2012-01-01	Active	5000000
Galaxy S11	Smartphone	Android	2011-01-01	Active	2000000
Galaxy S10	Smartphone	Android	2010-01-01	Active	1000000
Galaxy S9	Smartphone	Android	2009-01-01	Active	500000
Galaxy S8	Smartphone	Android	2008-01-01	Active	200000
Galaxy S7	Smartphone	Android	2007-01-01	Active	100000
Galaxy S6	Smartphone	Android	2006-01-01	Active	50000
Galaxy S5	Smartphone	Android	2005-01-01	Active	20000
Galaxy S4	Smartphone	Android	2004-01-01	Active	10000
Galaxy S3	Smartphone	Android	2003-01-01	Active	5000
Galaxy S2	Smartphone	Android	2002-01-01	Active	2000
Galaxy S1	Smartphone	Android	2001-01-01	Active	1000

- Use Data Extract instead of Live.
- Apply filters to reduce unnecessary data.
- Remove unused fields.
- Refresh the workbook.